MEAN ABSOLUTE DEVIATION AT MULTIPLE RESOLUTIONS

Summary

The Mean Absolute Deviation (MAD) measures how far on average deviations are from the Y=X line. Deviations are the observations of map Y minus the observations of map X. Quantity component is the difference in mean value between two images. Stratum compares the distribution of deviations across strata, and Element measures positive and negative values within a stratum.

Usage

This measure could be used to characterize differences between time points or used to validate model or sensor performance. This could include performing MAD analysis on predicted versus observed values, and measuring differences attributable to the method of data collection. In applications it will be necessary that both maps present the same continuous variable. For example, one could run this type of analysis on two precipitation maps, however they cannot perform it on maps of precipitation and temperature.

Syntax

**Inputs must be in an equal area projection. This program assumes that pixels are of the same height and width.*

Parameters	**Explanation**	**Example**
Variable Unit	MAD results will be given in the measurement of the observed variable.	"Degrees Celsius"
Image X	TIFF or GeoTIFF raster image to use as initial case.	SST_Feb1982.tif
Image Y	TIFF or GeoTIFF raster image to use as latter case.	SST_Aug1982.tif
Stratum	TIFF of GeoTIFF raster image that specifies strata or masked pixels.	SST_NSmask.tif
Original Resolution	The length of the pixel side. User will type the number and use the dropdown list for the unit of measure.	"100 Kilometers"
Coarsening Factor	The number of times greater the length of a coarse pixel side is compared to the length of the original resolution pixel side.	"5"
Workspace	Output folder path where results should be saved.	C:\MyMADfiles

Results

When MAD is done running, it will output the following:

- A scatter plot with pixels color-coded by Stratum with a Y=X line to show where values are the same for both X and Y images.
- A stacked bar graph of Quantity, Stratum, and Element components, adding to total Mean Absolute Deviation across multiple spatial resolutions.
- A csv file that gives the exact value for all components and Total MAD.

Methodology

Notation	Meaning
Xsg_{rc}	the X value in stratum s at grain g for the element in row r and column c.
Ysg_{rc}	the Y value in stratum s at grain g for the element in row r and column c.
Dsg_{rc}	the Ysg_{rc} minus Xsg_{rc} deviation in stratum s at grain g for the element in for row r and column c.
g	grain, or the number of times bigger the length of a coarse pixel side is compared to the length of the original resolution pixel side.
S	the number of strata.
N	the number of elements at the finest grain.
Rg	the number of rows at grain g.
Cg	the number of columns at grain g.

Dsg_{rc} = Ysg_{rc} - Xsg_{rc} (Eq.1)

Quantity Component = $QMAD = \frac{\left|\sum_{s=1}^{s} \sum_{r=1}^{Rg} \sum_{c=1}^{Cg} Ds1_{rc}\right|}{N}$ (Eq.2)

Stratum Component = $SMAD = \frac{\sum_{s=1}^{s}\left|\sum_{r=1}^{Rg} \sum_{c=1}^{Cg} Ds1_{rc}\right|}{N} - QMAD$ (Eq.3)

Element Component at grain g = $EMAD_g = \frac{\sum_{s=1}^{s} \sum_{r=1}^{Rg} \sum_{c=1}^{Cg} |Dsg_{rc}|}{N} - SMAD - QMAD$ (Eq.4)

Dsg_{rc} = $\sum_{i=g(r-1)+1}^{gr}\left[\sum_{j=g(c-1)}^{gc}(Y_{ij} - X_{ij})\right]$ (Eq.5)

$Rg = CEILING({}^{R1}/_{g})$ (Eq.6)

$Cg = CEILING({}^{C1}/_{g})$ (Eq.7)

$sgn = SIGN(\sum_{r=1}^{R1} \sum_{c=1}^{C1} Ds1_{rc})$ (Eq.8)

Equation 1 expresses deviation as the values of Y minus the values of X. Equation 2 computes the Quantity component, which is constant across resolutions. Equation 3 computes the Stratum Component, which is also constant across resolutions. If at least one $Dsg_{rc} > 0$ in one stratum and at least one other $Dsg_{rc} < 0$ in another stratum, then $SMAD > 0$; otherwise $SMAD = 0$. Equation 4 computes Element component at grain g. Data at the original resolution will be expressed with a grain of 1. If $Dsg_{rc} > 0$ in one pixel and $Dsg_{rc} < 0$ in another pixel, then $EMAD_g > 0$; otherwise $EMAD_g = 0$. The relation of these components is expressed as $QMAD \leq SMAD \leq EMAD_g$.

The inclusion of multiple resolution analysis will influence only element component. Equation 5 coarsens the spatial resolution of a deviation map by summing the deviations of pixels at the original resolution into coarser pixels, where each coarser pixel coincides with g columns and g rows of pixels at the original resolution. Equation 6 computes the number of rows at resolution g and equation 7 computes the number of columns at resolution g, where CEILING is a function that rounds up to the next highest integer. SIGN indicates either a positive or negative value. Equation 8 reveals the direction of quantity component. If the mean value of Y is greater than the mean value of X then sgn will be positive. If the mean value of Y is less than the mean value of X then sgn will be negative.

Characteristics of MAD

MAD has properties that make the approach easy to interpret and adopt in a wide variety of applications. MAD has the same units as X and Y, therefore MAD is interpretable in terms of the phenomenon that X and Y portray. MAD is bounded only by a lower limit of zero, which occurs if and only if X equals Y for all pixels. MAD has no upper limit because X and Y frequently have no limit, in which case there is no limit to the sizes of the deviations. MAD never leads to division by zero because the denominator of MAD is the number of pixels at the original resolution, which is always positive. When MAD compares X to Y we will get the same results as when MAD compares Y to X. Symmetry is desirable when it is not clear which variable should be defined as X and which variable should be defined as Y. Lastly, the influence of each deviation on MAD is proportional to the absolute value of the deviation, which is intuitive.

Contact

For comments or concerns regarding the MAD program, the developer, Rebecca Chapman, can be reached by email at RChapman@clarku.edu.

For academic publications in the field of Accuracy Assessment and Land Change Science, please visit the webpage of Robert Gilmore Pontius Jr, PhD at http://www.clarku.edu/~rpontius/.